



UNIVERSITY OF LAY ADVENTISTS OF KIGALI

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Faculty of Computing and Information Sciences

Course Code & Name: EWA408510 – E-Commerce and Web Application | Final Examination
(Project-Based)
Academic year **2025-2026**

Item	Details
Course Code & Name	EWA408510 – E-Commerce and Web Application
Instructor	Eric Maniraguha
Assessment Type	Individual Project
Duration	13 Days
Submission Period	21 June – 3 July 2026
Maximum Marks	40 Marks (+5 Bonus Marks)

Instructions

1. Attempt **all tasks/questions**.
 2. Read each requirement carefully before beginning your work.
 3. Ensure that all required information and supporting materials are correctly submitted.
 4. You may use AI tools to improve efficiency and support your work; however, all submitted work must be your own understanding and effort.
 5. Any form of plagiarism, academic dishonesty, or inability to clearly explain and defend your submitted work may result in a score of **zero (0)**.
 6. Ensure your submission is well-organized, complete, and submitted before the deadline.
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Project Scenario

A local business in Rwanda plans to expand its operations by offering products and services online. The business requires a modern e-commerce platform that enables customers to browse products, manage shopping carts, place orders, and track purchases efficiently.

As a web developer, your task is to design, develop, deploy, and present a complete E-Commerce Web Application that addresses the needs of an online business.

You may choose any suitable business domain, including but not limited to:

- Bookstore
- Fashion Store
- Electronics Shop
- Pharmacy
- Restaurant Ordering System
- Grocery Store
- Handicraft Marketplace
- Beauty and Cosmetics Store
- Agricultural Products Marketplace
- Any other approved e-commerce concept

Project Requirements

A. Functional Requirements

1. User Interface (UI)

The application should provide:

- Responsive and professional design
- Homepage with navigation menu
- Mobile-friendly user experience
- Consistent branding and layout

2. Product Management

The application should support:

- Product listing page
- Product details page
- Product categories
- Product search and filtering functionality (recommended)

3. Shopping Cart

The application should allow users to:

- Add products to cart
- Remove products from cart
- Update product quantities
- Calculate totals automatically

4. Checkout Process

The application should provide:

- Customer information collection
- Order summary review
- Form validation
- Order confirmation page

5. Database Integration

The system should store and manage:

- Product information
- Customer information
- Orders and transactions
- Relationships between entities

B. DevOps and Deployment Requirements

6. GitHub Repository (Mandatory)

Students must:

- Host the project on GitHub
- Maintain meaningful commit history
- Use proper project structure and documentation
- Include a complete README.md file

7. Deployment (Mandatory)

Students must:

- Deploy the application online
- Ensure the application remains accessible during evaluation
- Provide the deployment URL

8. CI/CD Pipeline (Mandatory)

Students must implement a Continuous Integration and Continuous Deployment (CI/CD) pipeline using GitHub Actions or an equivalent platform.

The pipeline should include:

- Automated build process
- Automated testing (where applicable)
- Automated deployment (preferred)

Evidence of successful workflow execution must be included in the report.

9. Docker Containerization (Mandatory)

Students must:

- Create a Dockerfile
- Use docker-compose.yml when multiple services are required
- Successfully build and run the application using Docker
- Include screenshots demonstrating successful execution

C. Security Requirements (Recommended)

Students are encouraged to implement:

- User authentication and authorization
- Password hashing
- Input validation

- Protection against common web vulnerabilities
- Secure database interactions

Deliverables

Each student must submit the following:

1. GitHub Repository Link
2. Live Deployment URL
3. Project Report (3–5 Pages)
4. Database Script/File (if applicable)
5. Screenshots of the Application
6. CI/CD Workflow Evidence
7. Docker Build and Execution Evidence

Project Report Structure

The report should contain the following sections:

1. Introduction
2. Problem Statement
3. Project Objectives
4. System Features
5. Technologies Used
6. System Architecture
7. Database Design
8. Screenshots of the Application
9. GitHub Repository Link
10. Deployment Link
11. CI/CD Implementation
12. Docker Implementation
13. Challenges Encountered
14. Future Enhancements
15. Conclusion

Assessment Criteria (40 Marks)

Component	Marks
User Interface (UI/UX)	5
Product Management	4
Shopping Cart Functionality	4
Checkout Process	4
Database Integration	5
GitHub Usage & Documentation	3
Deployment	3

CI/CD Pipeline	4
Docker Implementation	4
Presentation & Oral Defense	4
Total	40

Innovation Bonus (Up to 5 Marks)

Additional marks may be awarded for implementing innovative features such as:

- Payment Gateway Integration (Mobile Money, Stripe, PayPal, etc.)
- AI-Powered Product Recommendations
- Analytics Dashboard
- Advanced Security Features
- Multi-Vendor Marketplace Functionality
- Real-Time Notifications
- Progressive Web Application (PWA)
- Unique and Creative Business Ideas

Presentation and Oral Defense

During the evaluation, students will be required to:

- Demonstrate application functionality
- Explain system architecture
- Present database design
- Demonstrate GitHub repository usage
- Explain CI/CD workflow
- Demonstrate Docker deployment
- Explain major coding decisions
- Answer questions related to their implementation

Students must be able to explain all components of their project confidently.

Important Notes

- *The project must be developed as a Web Application. Desktop applications are not permitted.*
 - *Students may use any programming language, framework, database, or hosting platform.*
 - *The project must be completed and submitted within thirteen (13) days.*
 - *All submitted work must be original.*
 - *Evidence of plagiarism or inability to explain the submitted work may result in a score of zero (0).*
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Closing Remark

As you begin this final project, I encourage you to approach your work with honesty, discipline, and commitment.

The Bible reminds us in Colossians 3:23:

“Whatever you do, work at it with all your heart, as working for the Lord, not for human masters.”

This verse encourages us to perform every task with dedication, integrity, and excellence. Let your work reflect not only your technical skills but also your character and professionalism.

I wish you wisdom, creativity, and success as you complete this project and continue your journey in software development, e-commerce, and digital innovation.

Keep striving for excellence in all you do.

Eric Maniraguha

Instructor – E-Commerce and Web Application